Listen Before You Classify: A DSP Wake-Up Front End for Energy-Efficient and Reliable Sound Alerts on Wearables

Anonymous Authors
Anonymous Institution

***Abstract*—Wearable sound-awareness devices for Deaf and hard-of-hearing people run a neural audio classifier on every half second of microphone input, which drains the battery, and pretrained classifiers miss or confuse safety sounds. We place a classical signal-processing front end before the classifier (YAMNet): a high-pass filter, short-time band energies and two minimum-tracking noise floors, a fast and a slow one, wake the classifier only when a new sound starts; a Goertzel resonator bank with a temporal-pattern matcher recognises standard alarm tones, and an impulse-rhythm detector recognises door knocks. The front end costs 0.5 % of one classifier inference. On 8 h of test streams built from real events and real backgrounds in six environments never used for tuning, the classifier sleeps 94 % of the time in quiet audio at almost unchanged recall, while overall recall rises from 33 % to 56 %: fire alarms on standard tones from 29 % to 93 % and real door knocks and doorbells from 16 % to 23 % ($p = 0.016$). A diagnosis of the classifier's false alarms (87 % are sirens labelled as fire alarms) leads to a veto that removes 83 % of them. On naturally recorded audio the gate was awake for 94 % of 2 765 labelled events in 692 real home recordings, saves about 72 % of classifier compute over a real household's day (99 % in an empty home, 35 % with the TV on) and raised no false alert in 5.5 h. On held-out real alarm recordings the system finds 5–13 points more smoke alarms than the always-on classifier. We also show that synthetic, standard-compliant alarms misjudge both detectors, and characterise real alarms to explain why.**

***Index Terms*—sound event detection, wake-up detector, Goertzel algorithm, noise-floor tracking, energy efficiency, assistive technology**

I. Introduction

Wearable sound-awareness devices turn important environmental sounds, such as smoke alarms, sirens and door knocks, into vibration and visual alerts for Deaf and hard-of-hearing (DHH) people [1]. They typically run a pretrained neural classifier such as YAMNet [2] on every half second of audio. This has two costs. *Energy*: the classifier runs continuously although nothing new happens most of the time, and battery drain was a key constraint for on-watch sound recognition [1]. *Reliability*: a general-purpose classifier confuses related safety classes and can miss standardised signals such as the ISO 8201 temporal-three (T-3) smoke-alarm pattern [3].

Both problems have a classical signal-processing answer. Most of the day the acoustic scene is stationary, which a cheap onset detector can tell without a neural network; and alarms are *designed* signals, with a known pitch and rhythm that a matched detector can recognise. We ask: *can an adaptive DSP front end that wakes the classifier only when a new sound starts, and that detects alarm tones itself, cut classifier computation substantially without losing detections, in acoustic environments it was never tuned on?*

Our contributions are:

1) A training-free front end (Butterworth high-pass, short-time band energies, a fast and a slow minimum-tracking noise floor, a Goertzel resonator bank with a T-3 matcher, an impulse-rhythm knock detector) costing 0.5 % of one classifier inference.
2) A controlled comparison against continuous inference, duty cycling and a fixed energy gate, calibrated on 11 environments and tested on 6 unseen ones: the classifier sleeps 94 % of the time in quiet audio and 77 % in event streams with one of 288 detections lost (83 % and 67 % with none lost), with a classifier duty that is far more stable across rooms than a fixed threshold.
3) Higher recall at no extra false alarms: DSP detectors for the classes the classifier handles worst, standard alarm tones and door knocks, raise overall recall from 33 % to 56 %; a false-alarm diagnosis and a one-line veto remove 83 % of false alarms.
4) Validation on naturally recorded audio: 94 % of 2 765 real labelled events wake the gate, the compute saving over a real household's day is about 72 %, and held-out real alarm recordings show where the DSP detector helps (in noise) and where it cannot (bells, horns, non-T-3 alarms).
5) Evidence that synthetic, standard-compliant alarms overstate a DSP detector (90 % vs. 25–30 % on real recordings) and understate the classifier, with a characterisation of real alarms that explains the gap; and a negative result: requiring classifier agreement for DSP alarms costs recall without removing look-alike alarms. All data are public and the study is reproducible with one command.

II. Related Work

Sound awareness for DHH users. SoundWatch [1] classified environmental sounds on a smartwatch and compared watch-only, phone and cloud architectures, finding battery drain a key concern; ProtoSound [4] personalised recognition from few examples. Neither reduces *when* the classifier runs.

Wake-up detectors and cascades. Always-on systems commonly place a low-power detector before a costly stage [5]; keyword spotting wakes large models with small ones [6], and a recent cascade screens audio with a linear detector before a trimmed YAMNet for violence detection [7]. We use interpretable, training-free DSP gates whose threshold follows the room through noise-floor tracking, evaluate them on naturally recorded home audio, and add a second, decision-making DSP path for alarm tones.

Noise estimation and detection. Our floor tracker is a simplified form of minimum statistics [8]; the ordered-statistic CFAR threshold we test as an ablation comes from radar detection [9]. Sigtia et al. compared environmental sound classifiers by accuracy and computational cost, including smoke alarms recorded in homes [10].

III. Method

A. Signals and classifier

Audio is sampled at $f_s = 16\,\text{kHz}$, the classifier's native rate. The Nyquist frequency of 8 kHz covers the piezo smoke-alarm band (about 2.8–3.6 kHz) and the 520 Hz low-frequency alarm [11] with its odd harmonics up to the 13th. The classifier, YAMNet (MobileNetV1, 521 AudioSet classes), sees a 0.975 s window every 0.5 s and costs $M_Y = 70\,744\,064$ multiply-accumulates (MACs) per window, counted from its convolution and dense layers. Its scores are mapped to three alert categories (fire alarm, siren, door) by the maximum over category labels. An alert fires when the category score reaches 0.30 in N classified windows in a row ($N = 2$ for fire alarm and siren, 1 for door; a run survives skipped windows if classified windows are at most 2 s apart), with a 20 s cooldown and the most urgent category winning.

B. Gate path

High-pass filter. Butterworth, order 2, $f_c = 100\,\text{Hz}$ ($-3\,\text{dB}$), one biquad

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}, \tag{1}$$

with a double zero at $z = 1$ and two poles inside the unit circle (Fig. 6). It removes DC, rumble and handling noise below the lowest target component, so they cannot wake the classifier.

Band energies and noise floor. Each 20 ms frame ($N = 320$, Hann) is transformed with a 512-point FFT and its power summed in $B = 16$ log-spaced bands (100 Hz–7.6 kHz), smoothed with a one-pole low-pass ($\tau = 60\,\text{ms}$). With $E_b[t]$ the band level in dB, each band's floor follows the minimum with a bounded rise,

$$F_b[t] = \min\big(E_b[t],\, F_b[t-1] + \rho T\big), \qquad \rho = 3\ \text{dB/s}, \tag{2}$$

a simplified form of minimum statistics [8]: it drops to a quieter background at once and absorbs a steady sound within seconds. A sound that swells more slowly than ρ (a siren fading in) drags this floor up with it and never stands out, so a second floor $S_b[t]$ with the same recursion and $\rho_s = 0.3\,\text{dB/s}$ remembers how quiet the room was.

Wake-up. The novelty

$$\nu[t] = \max\Big(\max_b\big(E_b - F_b\big),\ \max_b\big(E_b - S_b\big) - \delta\Big)[t], \tag{3}$$

with $\delta = 6\,\text{dB}$, the largest rise of any band above its own floors, so the threshold is relative to the room (the offset δ was chosen on tuning data, Table III). Before each classifier window, if the maximum of ν over the 0.5 s just received exceeds Δ, the classifier runs, and stays awake for a hangover H. The window spans the last 0.975 s, so the onset that woke the gate lies inside the first classified window: waking adds no delay.

C. Alarm path

Goertzel resonator bank. The power of DFT bin k in a frame is computed by the Goertzel recursion [12]

$$s[n] = x[n] + 2\cos(\omega_k)\, s[n-1] - s[n-2], \tag{4}$$

$$|X_k|^2 = s_1^2 + s_2^2 - 2\cos(\omega_k)\, s_1 s_2, \tag{5}$$

a second-order IIR resonator with poles at $e^{\pm j\omega_k}$ on the unit circle, read once per frame. We use 17 probes from 2.8 to 3.6 kHz (50 Hz apart, one bin each) and 520, 1560 and 2600 Hz. A Goertzel probe needs one multiply per sample, so it beats a full $N \log_2 N$ FFT only below about 14 probes; with 20 probes, and the gate's FFT already computed, an implementation can read the probe bins from that FFT instead, halving the front end's cost. We keep the resonator formulation because it lets the alarm path run without the gate, for example on a low-power audio DSP.

Tone track and pattern. A frame is "on" when the strongest probe holds more than a share r of the frame energy (35 % for the three square-wave probes together) and is within 8 dB of its 1.5 s running peak, which rejects reverberant tails. After closing 100 ms dropouts, a fire alarm is declared on three pulses of 0.3–0.8 s separated by gaps of 0.3–0.8 s, the T-3 timing with tolerance. The first design (v3) used $r = 25\,\%$ and also required the 1.5 s pause after the third pulse ("strict"). The final design (v4) was selected on real recordings (Sec. IV-C): $r = 15\,\%$ without the pause check. Its alerts are merged with the classifier's under the same cooldown.

Knock path. A knock is a short broadband impulse. The onset strength is the mean over bands of the half-wave rectified rise of the band level since the previous frame (spectral flux in dB), computed from the gate's band energies. Impulses are local maxima above κ (at least 60 ms apart); a door alert is raised on m impulses with inter-onset intervals of 0.12–0.6 s and a regular rhythm (each interval within a factor 1.6 of the previous), when the classifier, if awake, gives the door classes a score of at least a within 2 s. κ, m and a were chosen on tuning data: most door recall with no more false alarms than without the detector ($\kappa = 15\,\text{dB}$, $m = 3$, $a = 0.05$).

Siren veto. A fire alarm from the classifier is accepted only when its score exceeds the siren score in the same window (motivated in Sec. V-C).

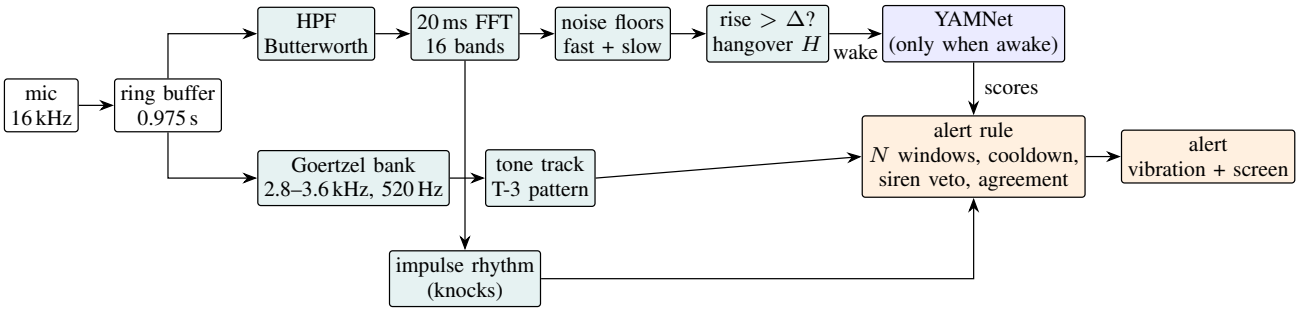


Fig. 1. The front end. The gate path wakes the classifier, which reads the last 0.975 s from the ring buffer; the alarm and knock paths raise alerts by themselves, merged with the classifier's under one alert rule.

D. Front ends and cost

F0 runs the classifier on every window; F1 on every k-th window; F2 wakes it when the high-passed level exceeds a fixed T dBFS; F3 is the adaptive gate; F4 is F3 plus the alarm path and the veto, and the proposed system is F4 plus the knock path. Per 0.5 s hop the gate costs about 196 000 MACs, the alarm path 177 500 and the knock path 1400, counted per stage, together 0.53 % of one classifier inference. Compute relative to F0 is $(d\,M_Y + M_{\text{gate}} + M_{\text{alarm}} + M_{\text{knock}})/M_Y$, with d the classifier duty (fraction of windows classified). We count multiply-accumulates rather than battery energy (Sec. VI).

IV. Experimental Setup

A. Test streams

Following the event-mixing protocol of DCASE 2017 Task 2 [13], we build 5-minute streams from DEMAND [14] recordings (8 microphones per environment as separate streams, background level -50 to -35 dBFS). Each event stream holds 6 targets (2 per category) at an SNR from $\{-5, 0, 5, 10, 20\}$ dB over the event's active frames, and 6 distractors (0–20 dB) that never overlap a target; each has an *idle* twin with the background only. Sirens, doorbells and knocks are real recordings from FSD50K [15] and ESC-50 [16] (official splits); distractors are other FSD50K/ESC-50 sounds, half of them easily confused with targets (bells, phones, doors, impacts, beeps, speech, music). Fire alarms are synthesised to ISO 8201 (piezo 2.9–3.5 kHz or a 520 Hz square wave, jittered timing and pitch, a random room response), because no public set of real smoke-alarm recordings exists. Tuning uses 11 environments (7.3 h of event and 7.3 h of idle streams, 528 targets); all reported numbers use 6 others (one per DEMAND category: living room, office, cafeteria, street, bus, park; 4.0 h + 4.0 h, 288 targets).

B. Calibration

For each gate we choose, on the tuning streams, the lowest-duty setting (Δ or T, and $H \in \{0, 1, 2\}$ s) whose *gate-caused* miss rate (the share of F0's detections the gated system loses) stays within a budget of 1, 3 or 5 %. Seven gate statistics were compared on tuning data at the deployed 5 % budget (Table III); the two-floor statistic with $\delta = 6$ dB had the lowest duty. Table I lists the chosen settings.

TABLE I
Settings chosen on the tuning environments.

Budget	1 %	3 %	5 %
F3: Δ (dB) / H (s)	13 / 0	18 / 0	22 / 0
F2: T (dBFS) / H (s)	-48 / 1	-44 / 2	-46 / 0

C. Real recordings

Alarms. 161 Freesound [17] recordings (CC0/CC-BY/CC-BY-NC): 38 smoke alarms, 30 fire alarms (bells, horns, beeps) and 93 look-alikes that should not raise a fire alarm (microwave, oven and kitchen timers, reversing beepers, alarm clocks, monitors, phones). They are split *by uploader* into a design half and a held-out half, so no recording device or room appears in both. The detector redesign searched probe band (4 widths), tone share (3 values) and pattern rule (T-3, strict T-3, periodic beeps, combinations) on the design half, with a rule fixed in advance: highest alarm recall with no fire alarm on design-half look-alikes. Each clip is tested clean and mixed into a test-environment background at 10 and 0 dB, after 5 s of that background so the noise floor has settled, as on an always-listening device.

Real scenes. DESED public evaluation set [18]: 692 real 10 s home recordings with time-stamped labels of 10 classes (2 765 events). SINS [19], [20]: real recordings of one home labelled with the household's activity; we score 1 981 10 s segments (5.5 h, up to 400 per activity), each after a 2 s warm-up. Nothing is tuned on these data.

D. Metrics

A target is detected if an alert of its category fires between its onset and 1 s after its end. We report recall with 95 % bootstrap intervals that resample whole streams (events in one stream share a background), false alarms per hour, median latency, classifier duty and relative compute; paired differences use the exact McNemar test.

V. Results

A. Compute and recall

Table II gives the test results. Duty cycling (F1) cannot save anything: classifying every second window already loses

TABLE II

TEST RESULTS (6 UNSEEN ENVIRONMENTS, 288 TARGETS). F4 = F3 + ALARM DSP + SIREN VETO; THE PROPOSED SYSTEM ADDS THE KNOCK DSP. FA/H ON EVENT STREAMS; DUTY IN BACKGROUND-ONLY STREAMS; COMPUTE RELATIVE TO F0.

Front end	Recall (%)	Fire (%)	Siren (%)	Door (%)	FA/h	Duty (%)	Comp.
F0 always on	32.6	29.2	53.1	15.6	11.5	100	1.00
F0 + veto	32.6	29.2	53.1	15.6	2.0	100	1.00
F0 + alarm DSP + veto	53.8	92.7	53.1	15.6	2.0	100	1.00
F1 every k-th	no $k > 1$ meets even the 5 % budget						
F2 fixed gate (5 %)	31.9	28.1	53.1	14.6	10.5	36	0.36
F3 two floors (5 %)	32.3	29.2	52.1	15.6	11.3	5.7	0.06
F4 (1 %)	53.8	92.7	53.1	15.6	2.0	49	0.49
F4 (3 %)	53.8	92.7	53.1	15.6	2.0	17	0.17
F4 (5 %)	53.5	92.7	52.1	15.6	2.0	5.7	0.06
F4 + knock (5 %)	**55.9**	**92.7**	52.1	**22.9**	**2.0**	**5.7**	**0.06**
F4, alarm + agreement	41.7	57.3	52.1	15.6	2.0	5.7	0.06

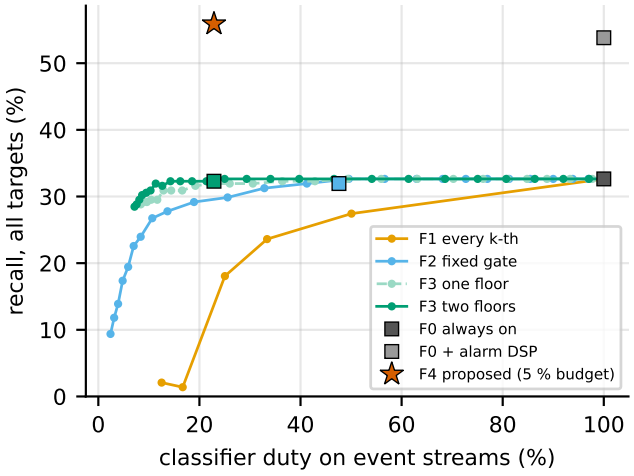


Fig. 2. Recall against classifier duty on the test event streams for duty cycling (F1), the fixed gate (F2) and the one- and two-floor gates (F3); markers: settings chosen on tuning data.

more than 5 % of detections on tuning data, so calibration returns $k = 1$. Both gates keep F0's recall within one or two of 288 detections. At the 5 % budget the two-floor gate lets the classifier sleep 94 % of the time in background-only audio and 77 % in event streams, against 64 % and 52 % for the fixed gate F2, at the same recall; at the 3 % budget it loses no detection at all and still sleeps 83 % of the time in quiet audio. Fig. 2 shows the full trade-off: the two-floor gate dominates the one-floor and the fixed gates, and duty cycling collapses.

The DSP detectors add recall where the classifier is weakest. The alarm path raises fire-alarm recall from 29 % to 93 % (synthetic standard tones, $p < 10^{-15}$); the knock path raises door recall (real knocks and doorbells) from 15.6 % to 22.9 % (7 more of 96, none lost, $p = 0.016$) with no extra false alarm. Together, recall rises from 32.6 % to 55.9 %.

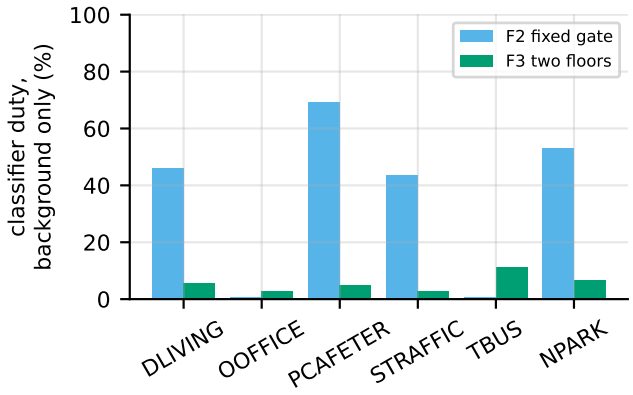


Fig. 3. Classifier duty in background-only audio per unseen test environment.

TABLE III

CLASSIFIER DUTY (%) ON THE TUNING ENVIRONMENTS WITHIN EACH GATE-CAUSED MISS BUDGET.

Gate	1 %	3 %	5 %
F1 every k-th	100	100	100
F2 fixed level	46.7	37.8	30.7
F3 one floor, max band rise	49.6	31.0	21.4
+ 150 ms smoothing, two bands	50.6	28.6	23.1
+ OS-CFAR	85.2	47.1	36.1
+ both	–	45.3	35.4
F3 two floors, $\delta = 6$ dB (selected)	**37.5**	**22.9**	**16.2**
$\delta = 10$ dB	39.8	28.5	19.9
$\delta = 14$ dB	49.7	28.1	21.5

B. Robustness across rooms

A fixed level threshold depends on how loud a room is. In background-only audio its duty at the 5 % budget ranges from 0.7 % (bus) to 69 % (cafeteria), while the two-floor gate stays between 2.7 % (office) and 11 % (bus) in all six unseen rooms (Fig. 3), because its threshold is relative to each band's own floors.

Gate statistic ablation. Longer smoothing with a two-band rule, and an ordered-statistic CFAR threshold (median novelty of the last 10 s) [9], did not improve on the one-floor statistic (Table III). CFAR was clearly worse: during a long sound such as a siren its reference median rises, and the statistic sinks just when the classifier's scores peak. The slow second floor helped at every budget: it wakes the classifier for slowly swelling sounds, which lets calibration raise Δ and keep the classifier asleep the rest of the time.

C. False alarms

All of F0's false alarms occur in event streams (none in 4 h of background): 40 of 46 are fire alarms raised during sirens, because YAMNet also scores *Alarm* on sirens and the fire alarm has priority. Tuning the classifier's thresholds on the tuning streams (best 0.275 instead of 0.30) does not change recall or false alarms. The siren veto cuts false alarms from 11.5 to 2.0 per hour (83 %) at unchanged recall and lowers the median alert latency from 1.85 to 1.55 s. Neither the DSP

TABLE IV
STRESS TEST (20 CLIPS EACH, 5 DB SNR): SHARE RAISING A FIRE ALARM (%). DSP DETECTORS: STRICT T-3, FINAL V4, AND V4 WITH CLASSIFIER AGREEMENT.

Signal	Fire?	YAMNet	Strict	v4	v4+agr.
T-3, 3.1 kHz (in spec)	yes	75	85	85	80
T-3, 520 Hz (in spec)	yes	0	90	90	45
T-3, 2.6 kHz / 4.0 kHz	yes	70 / 65	50 / 0	50 / 0	50 / 0
T-3, timing ×0.7 / ×1.5	yes	90 / 70	35 / 35	35 / 30	35 / 30
T-3, reverberation 1.5 s	yes	60	75	70	65
T-4 CO pattern	no	50	0	0	0
1 Hz beeper, 3.1 kHz	no	80	0	85	85
Single triple beep, 2.8 kHz	no	15	0	0	0

TABLE V
HELD-OUT REAL RECORDINGS (FREESOUND, UNSEEN UPLOADERS): SHARE RAISING A FIRE ALARM (%), CLEAN / 10 DB / 0 DB. "PROPOSED" IS THE FULL GATED SYSTEM.

Group (n)	YAMNet (F0)	DSP v4 alone	Proposed
Smoke alarms (23)	43 / 43 / 22	30 / 26 / 26	**52 / 48 / 35**
Fire bells, horns (13)	62 / 62 / 31	0 / 0 / 0	62 / 62 / 31
Look-alikes (50) ↓	12 / 6 / 4	2 / 2 / 0	14 / 8 / 4

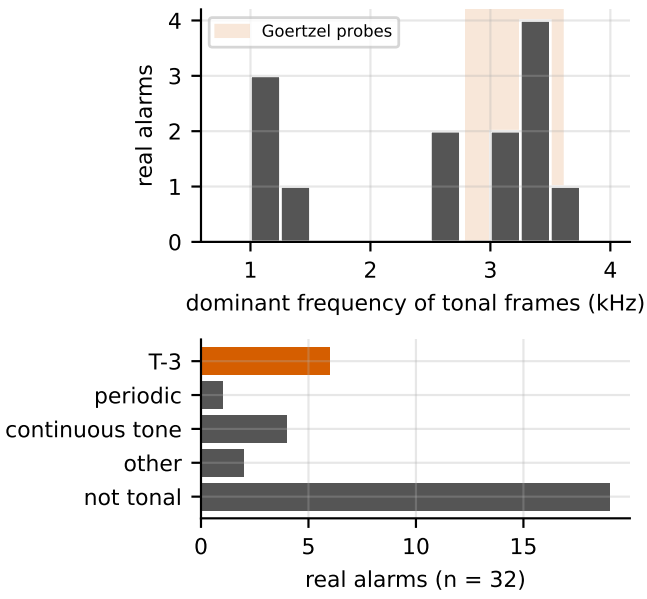


Fig. 4. Real alarms (design half, $n = 32$): dominant pitch of the tonal frames against the Goertzel probe band, and the pattern of each recording.

alarm path nor the knock path adds a false alarm on the 1 000+ everyday distractors.

Classifier agreement, a negative result. Following the intuition that a DSP fire alarm should be confirmed by the classifier (a fire score of at least 0.1 within 2 s), we tested it: it cut fire-alarm recall from 93 % to 57 % (it loses most 520 Hz alarms, which the classifier does not recognise) and did not stop the continuous beeper of Table IV, because the classifier itself scores *fire alarm* on beepers. We therefore do not use it.

D. Alarm detection: synthetic versus real

On the synthetic ISO 8201 alarms of the test streams, the DSP path raises fire-alarm recall from 29 % to 93 % ($p < 10^{-15}$); the classifier never detects the 520 Hz alarm, while the DSP path detects most of them (85 % in the first design). The gate does not delay alerts (median 1.84 s vs. 1.85 s for F0); the alarm path needs about 2.4 s to observe three pulses.

Real recordings tell a different story. Characterising the 32 design-half real alarms with DSP (FFT peak of tonal frames, beep timing) shows a median pitch of 3.0 kHz but only 54 % within the 2.8–3.6 kHz probes (range 1.0–3.5 kHz), and only 6 of 32 follow the T-3 pattern: 4 are continuous tones, 3 other rhythms, and 19 are not narrowband tones at all (bells, horns, distorted recordings). A tone-and-pattern detector can therefore only reach a minority of real alarms (Fig. 4).

The redesign raised design-half recall from 18 % to 25 % with no look-alike alarm; wider bands or a periodic-beep rule reached 27–33 % but fired on 2–17 % of look-alikes and were rejected.

On held-out real recordings (Table V) the classifier is much stronger than on synthetic alarms (43–62 % clean vs. 29 %) and the DSP detector much weaker (26–30 % vs. 93 %). Combined, the proposed system finds 5–13 points more smoke alarms than the always-on classifier, most at 0 dB, where the classifier degrades and the narrowband detector does not. The gate itself was awake during 98/91/77 % of the smoke alarms, and loses none of the fire bells, which a one-floor gate missed at 10 and 0 dB. The DSP detector fired on 1 of 50 held-out look-alikes, against 6 for the classifier.

E. Naturally recorded audio

On DESED's 692 real home recordings the gate was awake for 93.7 % of all labelled events (speech 97.3 %, alarm/bell/ringing 96.9 %, dishes 89.1 %, vacuum cleaner 84.4 %, the lowest, as a steady noise is partly absorbed by the floors), and the gated system kept 29 of the 32 alerts of the always-on classifier (Fig. 5); the one-floor gate, with a lower threshold, was awake for 96.9 % and kept 30. Over real home life the classifier duty follows the household: under 1 % in an empty home, 4 % while vacuuming, 6 % while working, 22 % while cooking and 65 % with the TV on. Weighted by how SINS divides the day, the classifier runs 27 % of the time, a 72 % compute saving including the front end. Neither the classifier (with the veto) nor the DSP paths raised an alert in the 5.5 h.

VI. DISCUSSION

What the front end buys. Compute and accuracy together: the same classifier detections at a fraction of its compute, more detections of the classes it handles worst, fewer false alarms, and no added delay from waking. The saving depends on the scene and should be reported per activity, not as one number: our mixed scenes suggest 94 % in quiet audio, a real household 72 %, an empty home 99 %, a home with the TV on 35 %.

Where DSP detection helps, and where it cannot. A matched DSP detector is robust in noise and rarely fooled by

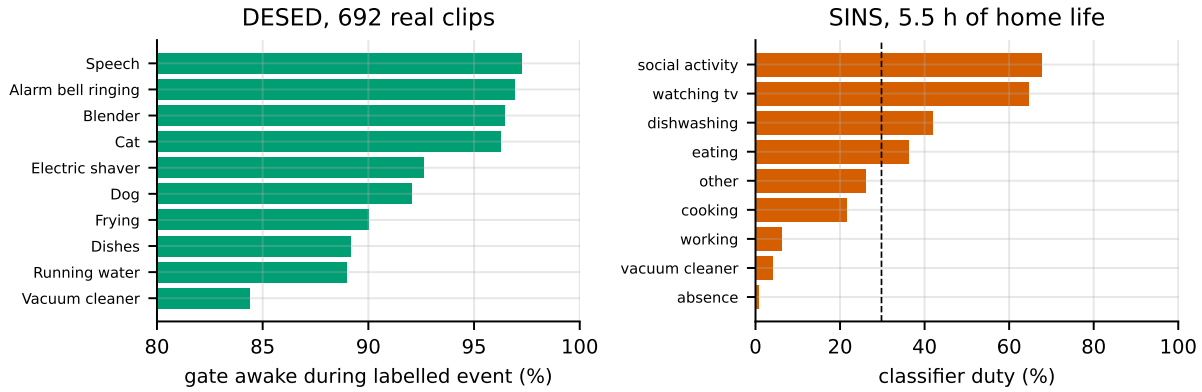


Fig. 5. Naturally recorded audio. Left: share of DESED's labelled events during which the gate was awake. Right: classifier duty of the proposed system per SINS household activity (dashed: all 5.5 h; weighted by the activities' share of the full SINS recordings, 27 %).

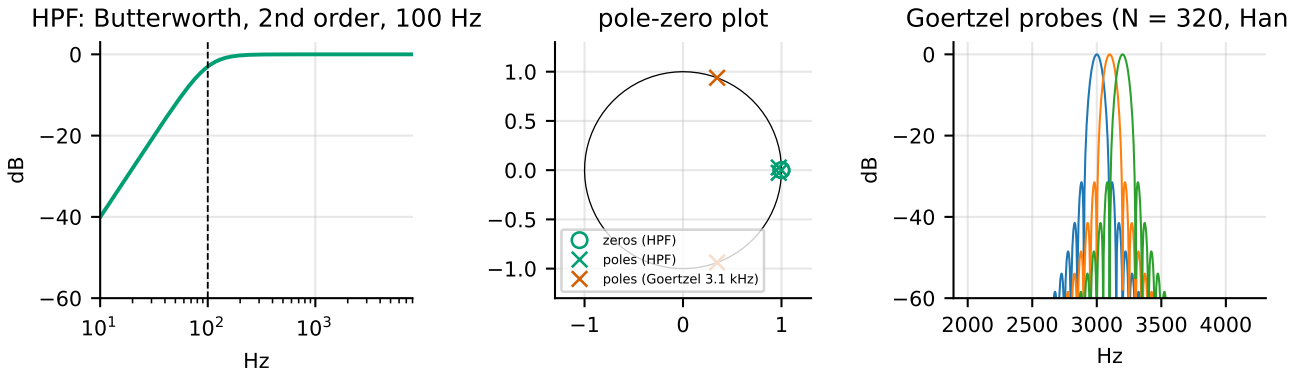


Fig. 6. DSP design: high-pass filter magnitude response, pole-zero plot (HPF and one Goertzel resonator) and Goertzel probe responses ($N = 320$, Hann).

look-alikes, but it only reaches alarms that are the narrowband patterns it encodes. On real recordings that is a minority, so it is a safety net beside the classifier, not a replacement. Synthetic, standard-compliant test alarms hide this: they made the DSP detector look three times better and the classifier half as good as on real recordings. Detector designs should be validated on real recordings, split by recording source.

Limitations. Real smoke-alarm recordings are scarce: 38 in total, so the held-out differences (2–3 clips) are not statistically significant. SINS segments were scored independently after a 2 s warm-up, not as continuous episodes, so the per-activity duty is an estimate; with $\rho = 3$ dB/s the fast floor needs about 7 s to absorb a 20 dB step such as a TV switched on, during which the classifier stays awake. We count multiply-accumulates, not battery energy: on a wearable the microphone, analog-to-digital conversion and the front end's own FFT run continuously, so the battery saving is smaller than the classifier saving and depends on how much of the device's power the classifier takes. The knock path gains 7 of 96 knocks; door recall stays low because the classifier itself rarely recognises knocks, even at 20 dB. The rule without the pause check (v4) fires on continuous 1 Hz beepers (Table IV); the strict rule does not; the first, strict design reached 7 points less recall on the design-half real alarms, and the choice depends on the deployment. Fire bells and horns are outside the detector's design. Results are for one classifier.

VII. Conclusion

A classical front end costing 0.5 % of one neural inference lets a wearable's sound classifier sleep most of the time without losing detections and adapts to unseen rooms where a fixed threshold does not; with matched alarm and knock detectors and a siren veto it raises recall from 33 % to 56 % and removes most false alarms. Validation on naturally recorded audio confirmed the gate and a 72 % compute saving over a real household's day, and showed that real alarms, unlike synthetic ones, are mostly beyond a tone-and-pattern detector. Future work: continuous real-home recordings with labelled safety events, on-device energy measurement, and detectors for the alarm types characterised here.

References

[1] D. Jain, H. Ngo, P. Patel, S. Goodman, L. Findlater, and J. Froehlich, "SoundWatch: Exploring smartwatch-based deep learning approaches to support sound awareness for deaf and hard of hearing users," in *Proc. ACM SIGACCESS Conf. Computers and Accessibility (ASSETS)*, 2020.

[2] S. Hershey *et al.*, "CNN architectures for large-scale audio classification," in *Proc. IEEE ICASSP*, 2017, pp. 131–135.

[3] "ISO 8201:2017 alarm systems – audible emergency evacuation signal – requirements," International Organization for Standardization, 2017.

[4] D. Jain, K. H. A. Nguyen, S. Goodman, R. Grossman-Kahn, H. Ngo, A. Kusupati, R. Du, A. Olwal, L. Findlater, and J. Froehlich, "ProtoSound: A personalized and scalable sound recognition system for deaf and hard-of-hearing users," in *Proc. CHI Conf. Human Factors in Computing Systems*, 2022.

[5] M. Gazivoda, D. Oletić, and V. Bilas, "Features and always-on wake-up detectors for sparse acoustic event detection," *Electronics*, vol. 11, no. 3, p. 478, 2022.

[6] A. Gruenstein, R. Alvarez, C. Thornton, and M. Ghodrat, "A cascade architecture for keyword spotting on mobile devices," 2017, arXiv:1712.03603.

[7] F. Zhu-Zhou, D. Tejera-Berengue, R. Gil-Pita, and M. Rosa-Zurera, "Acoustic violence detection using cascade strategy for computationally constrained scenarios," *Electronics*, vol. 15, no. 6, p. 1227, 2026.

[8] R. Martin, "Noise power spectral density estimation based on optimal smoothing and minimum statistics," *IEEE Trans. Speech Audio Process.*, vol. 9, no. 5, pp. 504–512, 2001.

[9] H. Rohling, "Radar CFAR thresholding in clutter and multiple target situations," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-19, no. 4, pp. 608–621, 1983.

[10] S. Sigtia, A. M. Stark, S. Krstulović, and M. D. Plumbley, "Automatic environmental sound recognition: Performance versus computational cost," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 24, no. 11, pp. 2096–2107, 2016.

[11] "NFPA 72: National fire alarm and signaling code," National Fire Protection Association, 2022.

[12] G. Goertzel, "An algorithm for the evaluation of finite trigonometric series," *Amer. Math. Monthly*, vol. 65, no. 1, pp. 34–35, 1958.

[13] A. Mesaros, T. Heittola, A. Diment, B. Elizalde, A. Shah, E. Vincent, B. Raj, and T. Virtanen, "DCASE 2017 challenge setup: Tasks, datasets and baseline system," in *Proc. DCASE Workshop*, 2017.

[14] J. Thiemann, N. Ito, and E. Vincent, "The diverse environments multi-channel acoustic noise database (DEMAND)," in *Proc. Meetings on Acoustics*, vol. 19, 2013.

[15] E. Fonseca, X. Favory, J. Pons, F. Font, and X. Serra, "FSD50K: An open dataset of human-labeled sound events," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 30, pp. 829–852, 2022.

[16] K. J. Piczak, "ESC: Dataset for environmental sound classification," in *Proc. ACM Int. Conf. Multimedia*, 2015, pp. 1015–1018.

[17] F. Font, G. Roma, and X. Serra, "Freesound technical demo," in *Proc. ACM Int. Conf. Multimedia*, 2013, pp. 411–412.

[18] N. Turpault, R. Serizel, A. P. Shah, and J. Salamon, "Sound event detection in domestic environments with weakly labeled data and soundscape synthesis," in *Proc. DCASE Workshop*, 2019.

[19] G. Dekkers, S. Lauwereins, B. Thoen, M. W. Adhana, H. Brouckxon, B. Van den Bergh, T. van Waterschoot, B. Vanrumste, M. Verhelst, and P. Karsmakers, "The SINS database for detection of daily activities in a home environment using an acoustic sensor network," in *Proc. DCASE Workshop*, 2017.

[20] G. Dekkers, L. Vuegen, T. van Waterschoot, B. Vanrumste, and P. Karsmakers, "DCASE 2018 challenge – task 5: Monitoring of domestic activities based on multi-channel acoustics," 2018, arXiv:1807.11246.